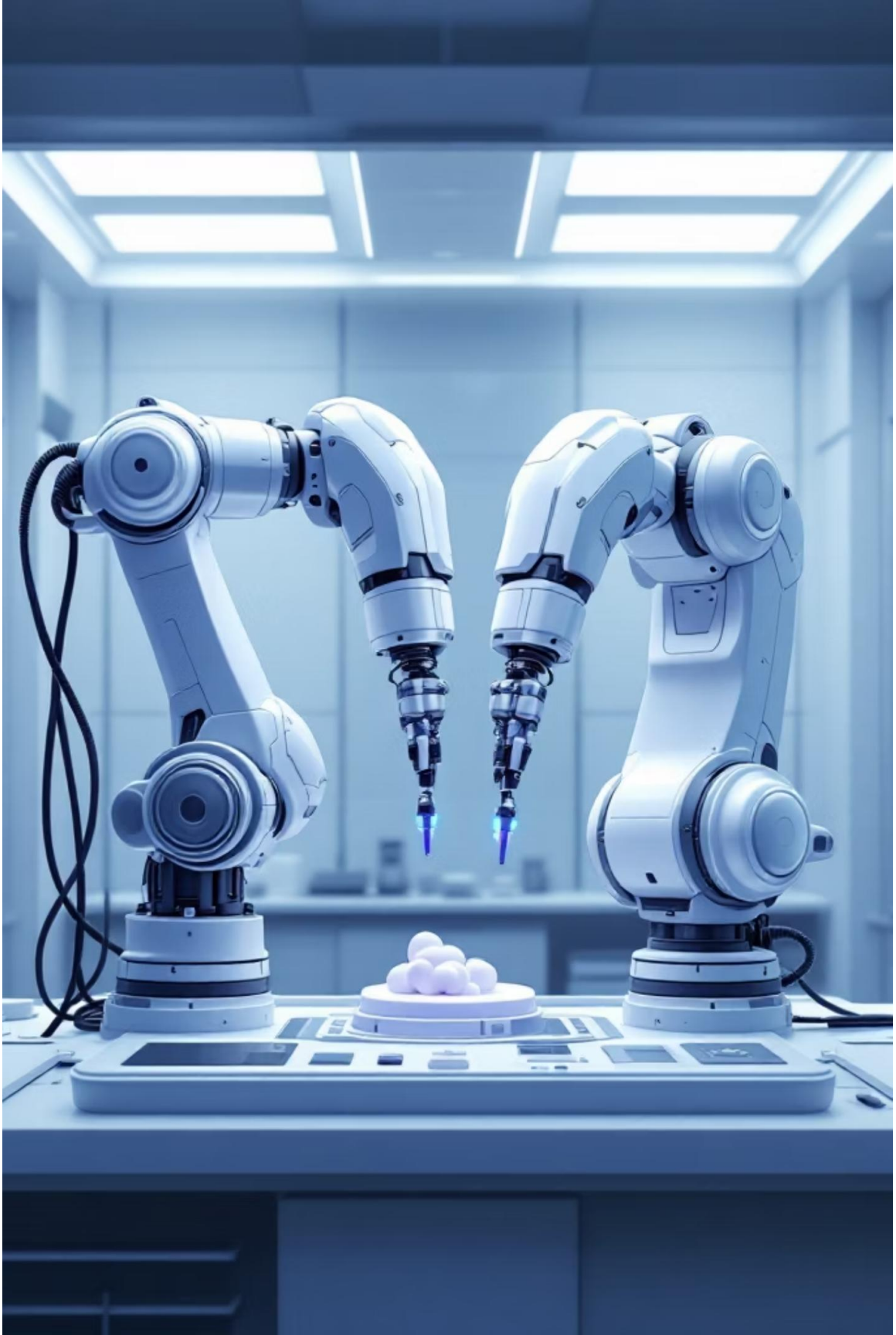


Diffusion Policy Imitation Learning for Dual- Arm Manipulation

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Core Challenges :

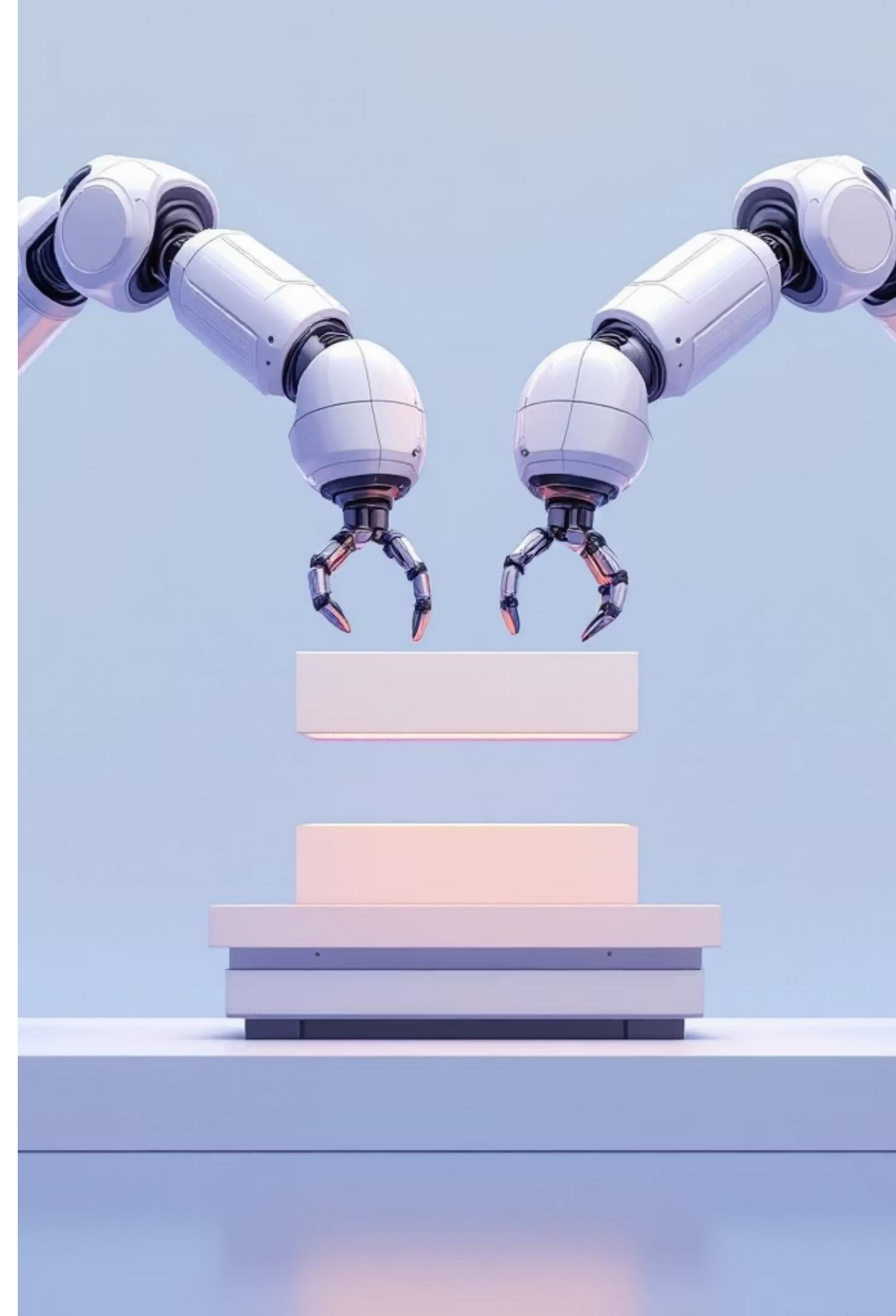
Why is Dual-Arm Manipulation a Litmus Test for Embodied AI?

- Embodied AI requires a closed perception-action loop — unlike internet-scale AI

Major Difficulties

- High-dimensional action spaces (14-DoF)
- Non-convex action distributions
- Sensitive closed-chain contact dynamics

We therefore systematically evaluate IL algorithms, starting with the simplest: Behavioral Cloning



RoboSuite Simulation & Expert Data Collection

- **Environment:** TwoArmLift (MuJoCo) – two Sawyer arms cooperatively lift a rigid pot.
- **State space (50-dim):** joint positions/velocities, 6D end-effector pose (quaternions), grasp state, relative object coordinates.
- **Action space (14-dim):** delta-position + gripper open/close, range $[-1,1]$.
- **Expert data collection:** Keyboard teleoperation, object position randomized ($\pm 0.15\text{m}$) $\rightarrow$ hundreds of trajectories.
- **Figure suggestion:** Simple diagram showing state/action space, or data collection setup.
- **Key insight:** Custom human-in-the-loop pipeline built due to scarcity of public dual-arm datasets.
- **Takeaway:** Dataset has initial-position diversity but limited size – IL must be sample-efficient.
- **Transition:** First, test the simplest method – Behavioral Cloning. IL algorithms, starting with the simplest: Behavioral Cloning

BC: Theoretical Defects & Empirical Failure

Theoretical problem

Violates i.i.d. assumption

→ compounding errors

→ $O(T^2)$ divergence

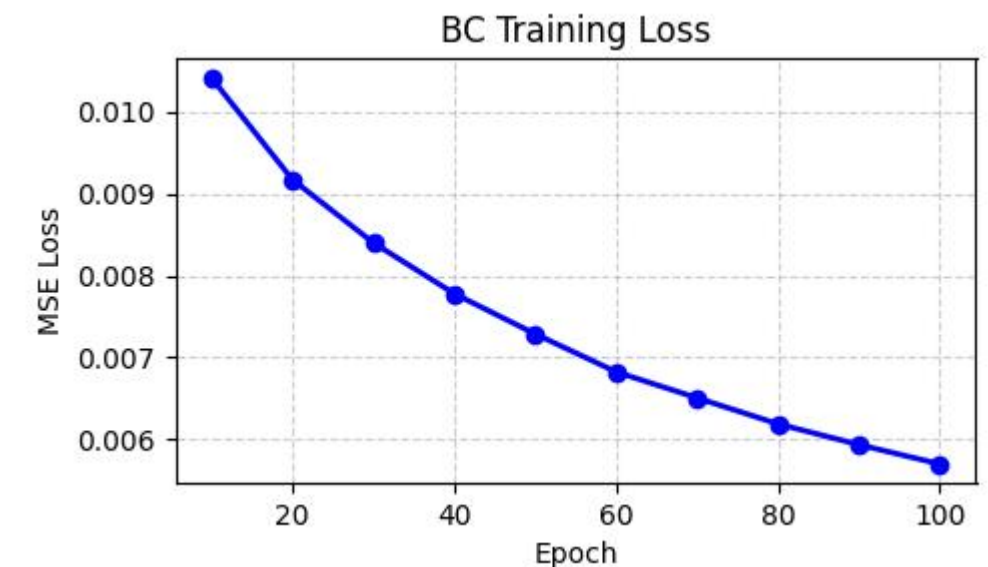
→ covariate shift

Experiment

Train on expert data, test under random initial positions

Result

0% success rate – low training loss but complete closed-loop failure



- Low training loss does not imply robustness to unseen state distributions
- BC completely fails in this dual-arm task.

DAgger: Better in Theory, Constrained in Practice

Limitation

Requires a continuously queryable expert (feasible in simulation, extremely hard on real robots)

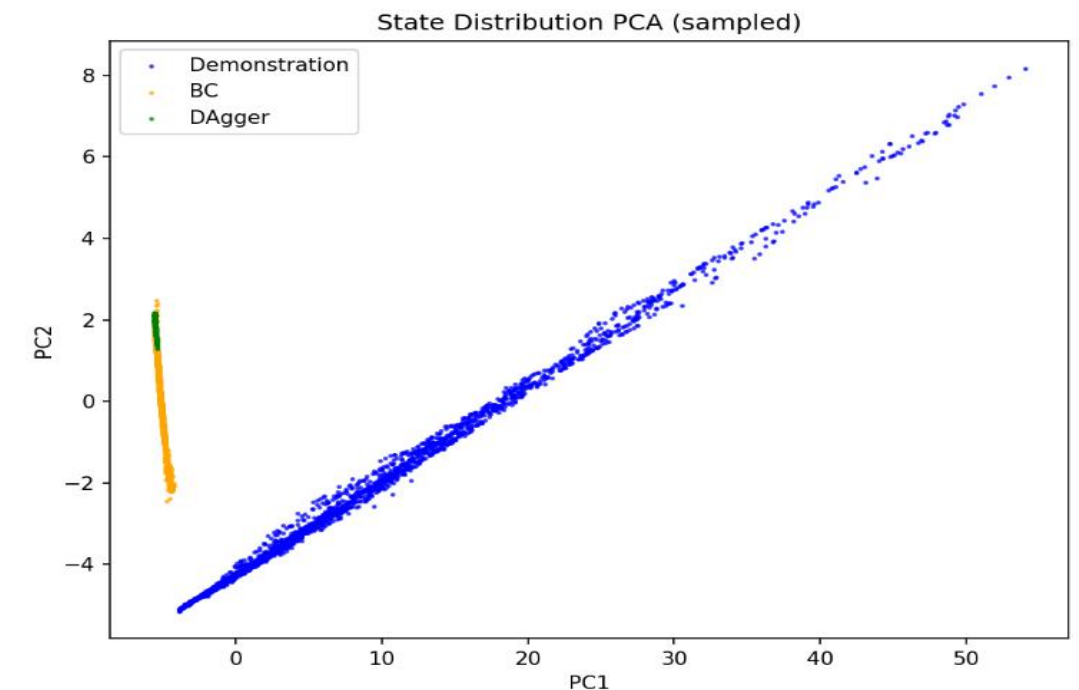
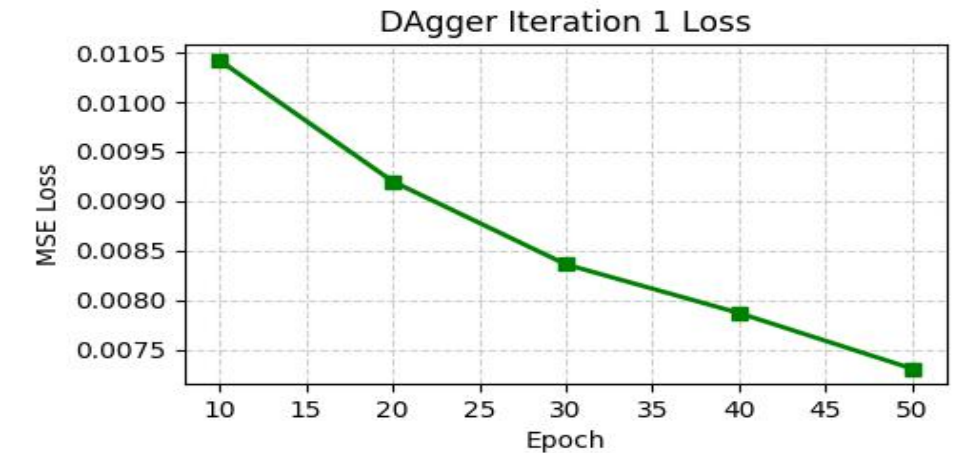
Our implementation

Multi-round data aggregation within simulation

Result

Modest improvement over BC, but still fails during contact-sensitive phases
→ 0% success rate

- Even with expert interventions, covariate shift persists. Need a method that explicitly models multi-modal distributions



Proposed Methodology: Diffusion Policy

Diffusion Policy theoretically overcomes BC/Dagger limitations. But theory alone is not enough – we need engineering optimizations to make it converge.

01 Forward process

Gradually add noise to expert action sequences → pure noise

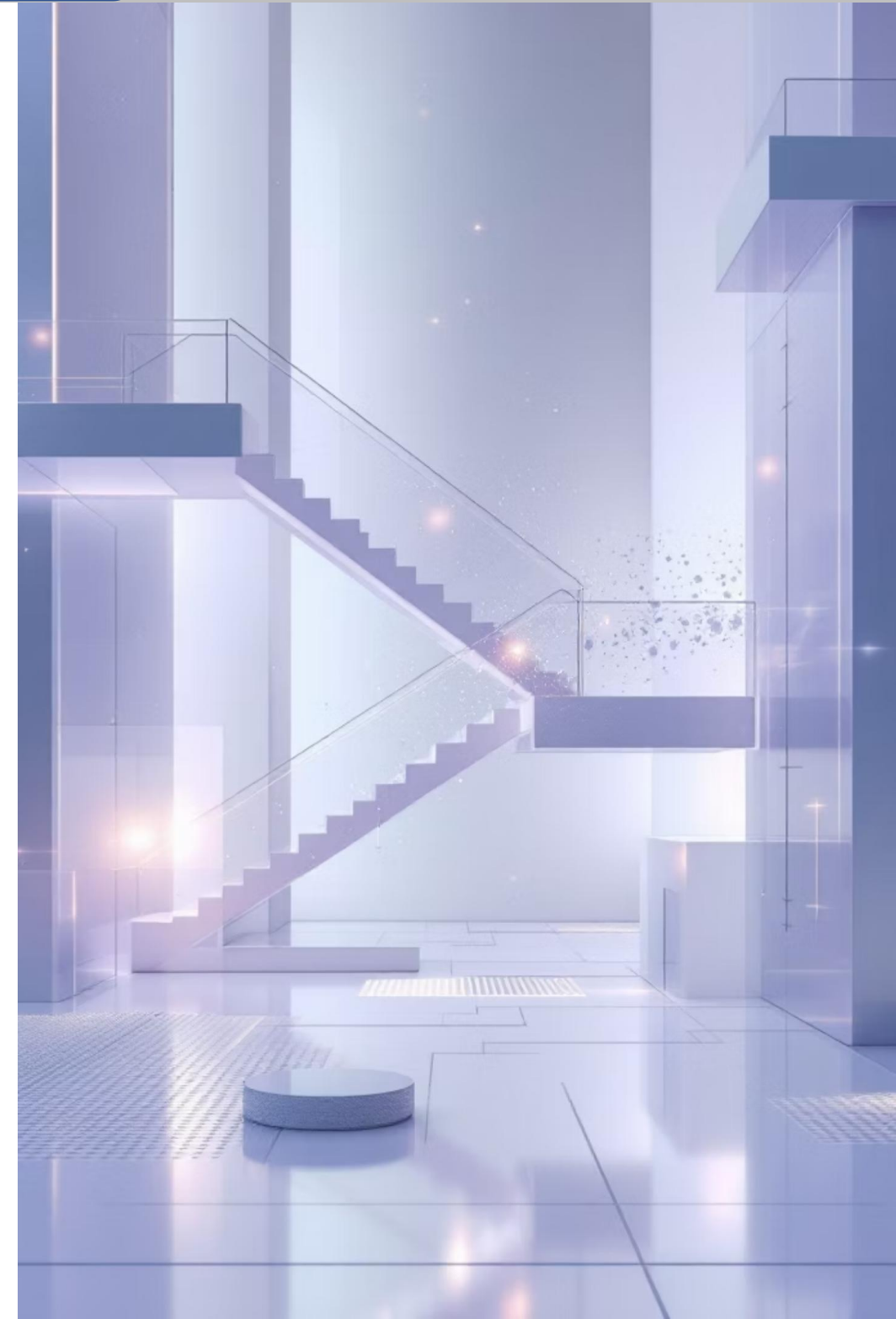
02 Reverse process

Learn to denoise – recover action sequences from noise

03 Conditioning

Observation O injected via FiLM layers in a 1D-UNet

$$x_{k-1} = \frac{1}{\sqrt{\alpha_k}} \left(x_k - \frac{1 - \alpha_k}{\sqrt{1 - \bar{\alpha}_k}} \epsilon_{\theta}(x_k, k, O) \right) + \sigma_k z$$



Architecture: 1D-UNet & Action Chunking

Model Backbone

Conditional 1D-UNet with Feature-wise Linear Modulation (FiLM) for state conditioning.

Temporal Strategy

Implementing Receding Horizon Control—predicting 16 action steps but only executing the first 8 to maintain smoothness.

Proprioceptive Input

Processing a 50-dimensional state vector (Joint positions, EEf poses, and object coordinates).



Training Engineering & Stability Optimizations



EMA Strategy

Using Exponential Moving Average with a power-based warmup to stabilize weights.



Normalization

Replacing BatchNorm with GroupNorm to ensure compatibility with small batch sizes and EMA.



Convergence Result

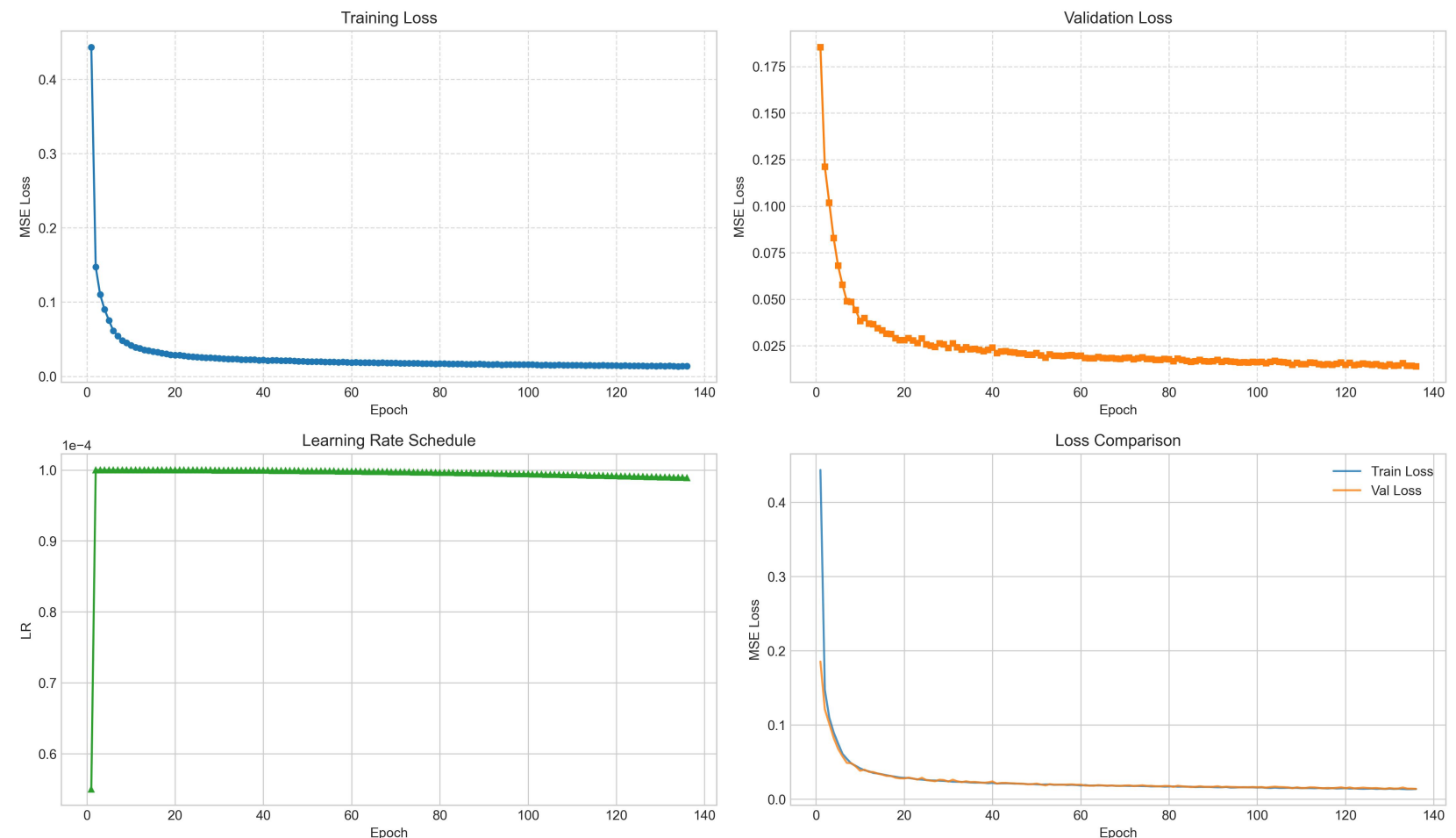
Demonstrating near-perfect training loss reduction and high-fidelity distribution matching.

Training Convergence Results

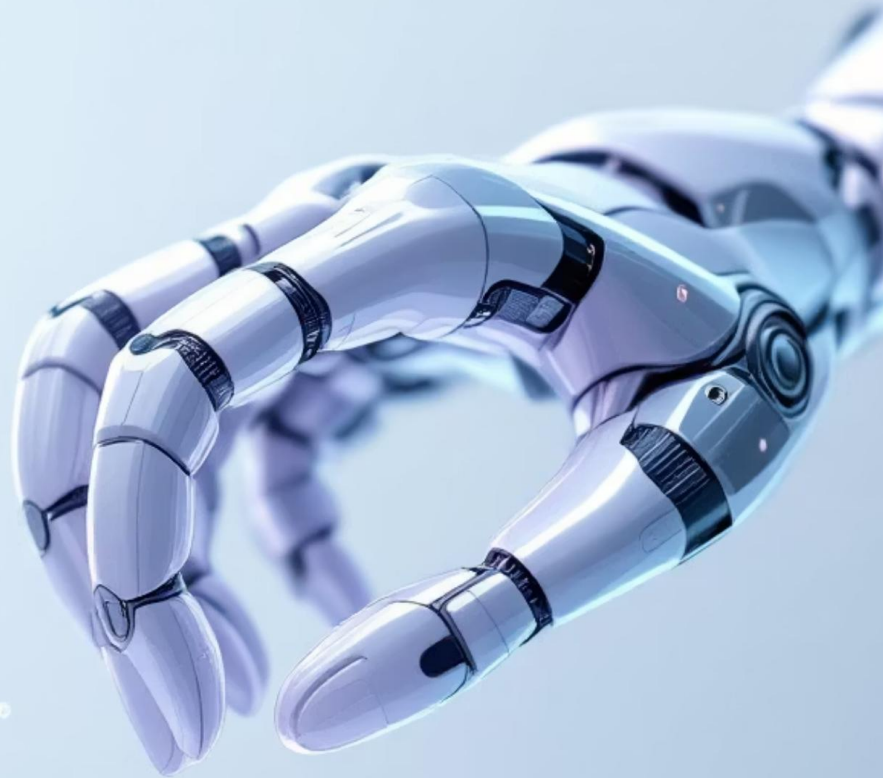
- Trained for 1500 epochs – denoising MSE loss quickly drops and plateaus at ~ 0.005 .
- No severe oscillations – EMA and gradient clipping ($\text{max_norm}=1.0$) work well.

From a supervised learning perspective,
Diffusion Policy is a success.

Diffusion Policy Training Metrics



However, when deployed in closed-loop simulation, the result is shocking !



Diagnosis: The Convergence-Execution Gap

Finding 1: Action Rigidity

Open-loop action chunking provides smoothness but lacks "reactivity" during the 8-step execution window.

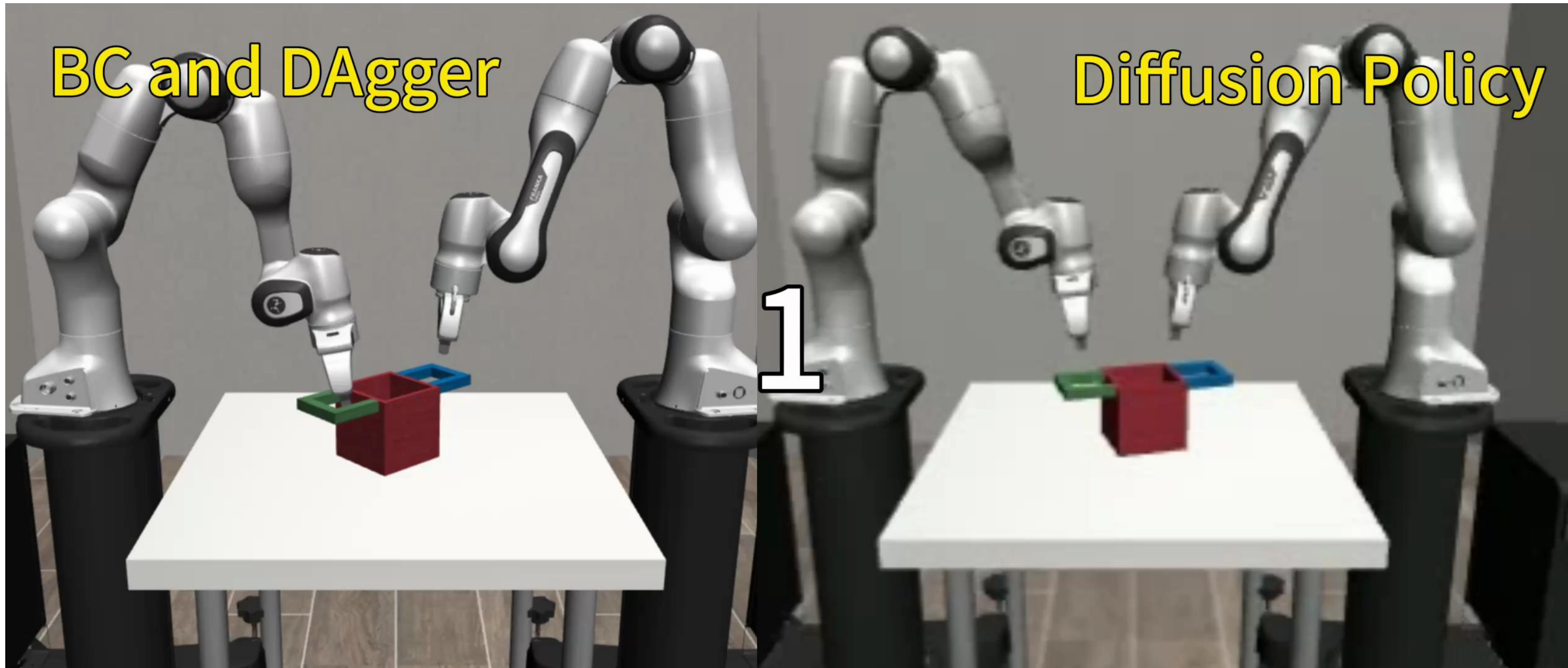
Finding 2: Contact Sensitivity

The MuJoCo physics engine is unforgiving; millimeter-scale errors in the state-only modality lead to object slippage.

The Paradox

Statistical perfection in training does not equal physical robustness in contact-rich tasks.

Demonstration Video



Why does a perfectly converged model
fail in the physical loop?

Failure Mode

Open-Loop Chunking Rigidity

- Policy generates 16-step chunk, then open-loop executes first 8 steps.
- Works during reach (no contact) but fails during lift (contact-rich).
- At contact, MuJoCo introduces tiny non-linear resistances – open-loop actions cannot adjust → pot pushed over or slips.
- **Need shorter action chunks or closed-loop correction mechanisms**

Limits of Proprioceptive Modality

- Current model uses only 50-dim proprioceptive state (joint angles, end-effector poses, etc.).
- During grasp & lift, the policy needs precise spatial awareness of the pot's handles and contact geometry.
- No explicit geometry → model often applies asymmetric forces → closed kinematic chain breaks → object slips.
- **Vision is essential to provide spatial attention and geometric reasoning.**

Future Directions



Vision+State pipeline

Integrate frontview and agentview (cropped to 216×288), process via ResNet/ViT, fuse with state vector. → Enables explicit geometry perception, symmetric force control.



Inference acceleration

Current DDPM requires 100 denoising steps → high latency.
→ Switch to DDIM, reduce to 10–20 steps → double control frequency, improve closed-loop reactivity.



RL Fine-Tuning

Diffusion strategy as a powerful behavioral prior. On the basis of this prior, the PPO algorithm is fine-tuned with dense reward functions (such as proximity to target and contact force symmetry). Agent can actively explore failure boundaries and learn recovery actions not present in static dataset.

Conclusion

- **Systematically evaluated BC → DAgger → Diffusion Policy on a dual-arm task.**
- **Diffusion Policy achieves perfect training convergence but 0% closed-loop success – the Convergence-Execution Gap.**
- **Root causes: open-loop action chunking + lack of visual perception.**
- **Future must combine vision-state fusion, fast inference (DDIM), and RL fine-tuning.**



Thank you!

Project Code - GitHub

- <https://github.com/joeypeng1023/Diffusion-Policy-Imitation-Learning-for-Dual-Arm-Manipulation>